

Java User Input

Mastering the Scanner Class

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General Introduction

User interaction is a core part of many applications. In Java, the primary way to capture user input from the console is through the **Scanner** class.

The Scanner class is part of the `java.util` package. It can parse primitive types and strings using regular expressions.

Requirement: You must always import the Scanner class before using it:

```
import java.util.Scanner;
```

1. Initializing the Scanner

To use the Scanner, you create an object of the class and pass `System.in` (standard input stream) to its constructor.

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        // Create the object
        Scanner myObj = new Scanner(System.in);

        System.out.println("Enter username:");

        // Read a line of text
        String userName = myObj.nextLine();

        System.out.println("Username is: " + userName);
    }
}
```

Expected Behavior: The program pauses and waits for the user to type something and press *Enter*.

2. Dynamic Input Types

The Scanner class provides specialized methods to read various data types directly without manual conversion.

Method	Input Description
<code>nextBoolean()</code>	Reads a true/false value
<code>nextByte()</code>	Reads a byte value
<code>nextDouble()</code>	Reads a fractional double
<code>nextFloat()</code>	Reads a single-precision float
<code>nextInt()</code>	Reads a whole integer
<code>nextLine()</code>	Reads a full String (whole line)
<code>nextLong()</code>	Reads a long integer
<code>nextShort()</code>	Reads a short integer

3. Reading Multiple Inputs

You can use the same Scanner object to read different types of data sequentially. This is useful for forms or user profiles.

```
System.out.println("Enter name, age, and salary:");

// String Input
String name = myObj.nextLine();

// Numerical Input
int age = myObj.nextInt();
double salary = myObj.nextDouble();

System.out.println("Profile: " + name + " | Age: " + age);
```

Warning: InputMismatchException

If you provide a value that doesn't match the expected type (e.g., typing "Hello" when `nextInt()` is called), the program will crash with an Exception.

Summary Checklist

Keep these rules in mind when implementing user input in your projects:

- ✓ Always import `java.util.Scanner` at the top.
- ✓ Use `System.in` as the input source.
- ✓ Choose the method that matches your data type (e.g., `nextInt` for numbers).
- ✓ Use `nextLine()` to capture spaces in strings.
- ✓ Close your scanner if you are warned about resource leaks (optional for `System.in`).

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